Exam Date & Time: 24-Dec-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: HUM 3021/HUM_3021 - ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT (MAKEUP)

ENGINEERING ECONOMICS AND FINANCIAL MANAGEMENT [HUM 3021]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Missing data may be suitably assumed.
Interest factor table is provided for others use formula.
Formula book and Interest factor table are included in reference material section.

1A) Suppose you have borrowed ₹1,00,000 at an interest rate of 8% per year compounded semiannually and desire to repay the money with six equal end-of-year payments, with the first payment made three years after receiving the ₹1,00,000.
i) Estimate the size of the annual payment. (4)
ii) At the time of the fifth payment, suppose you decide to pay off the loan with one lump sum payment. How much should it be paid?

1B) Alekhya is a 22-year-old senior who used the university's loan program to borrow ₹4,000 four years ago when the interest rate was 4.06% per year. ₹5,000 was borrowed three years ago at 3.42%. Two years ago, she borrowed ₹6,000 at 5.23%, and last year ₹7,000 was borrowed at 6.03% per year. Now she would like to consolidate her debt into a single 20-year loan with a 6% fixed annual interest rate. If Ashea makes annual payments (starting in one year) to repay her total debt, what is the amount of each payment? (3)

6% Compound Interest Factors 6%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
1	1.060	.9434	1.0000	1.0600	1.000	0.943	0	0	1
2	1.124	.8900	.4854	.5454	2.060	1.833	0.485	0.890	2
3	1.191	.8396	.3141	.3741	3.184	2.673	0.961	2.569	3
4	1.262	.7921	.2286	.2886	4.375	3.465	1.427	4.945	4
5	1.338	.7473	.1774	.2374	5.637	4.212	1.884	7.934	5
6	1.419	.7050	.1434	.2034	6.975	4.917	2.330	11.459	6
7	1.504	.6651	.1191	.1791	8.394	5.582	2.768	15.450	7
8	1.594	.6274	.1010	.1610	9.897	6.210	3.195	19.841	8
9	1.689	.5919	.0870	.1470	11.491	6.802	3.613	24.577	9
10	1.791	.5584	.0759	.1359	13.181	7.360	4.022	29.602	10
11	1.898	.5268	.0668	.1268	14.972	7.887	4.421	34.870	11
12	2.012	.4970	.0593	.1193	16.870	8.384	4.811	40.337	12
13	2.133	.4688	.0530	.1130	18.882	8.853	5.192	45.963	13
14	2.261	.4423	.0476	.1076	21.015	9.295	5.564	51.713	14
15	2.397	.4173	.0430	.1030	23.276	9.712	5.926	57.554	15

6% Compound Interest Factors 6%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
16	2.540	.3936	.0390	.0990	25.672	10.106	6.279	63.459	16
17	2.693	.3714	.0354	.0954	28.213	10.477	6.624	69.401	17
18	2.854	.3503	.0324	.0924	30.906	10.828	6.960	75.357	18
19	3.026	.3305	.0296	.0896	33.760	11.158	7.287	81.306	19
20	3.207	.3118	.0272	.0872	36.786	11.470	7.605	87.230	20
21	3.400	.2942	.0250	.0850	39.993	11.764	7.915	93.113	21
22	3.604	.2775	.0230	.0830	43.392	12.042	8.217	98.941	22
23	3.820	.2618	.0213	.0813	46.996	12.303	8.510	104.700	23
24	4.049	.2470	.0197	.0797	50.815	12.550	8.795	110.381	24
25	4.292	.2330	.0182	.0782	54.864	12.783	9.072	115.973	25
26	4.549	.2198	.0169	.0769	59.156	13.003	9.341	121.468	26
27	4.822	.2074	.0157	.0757	63.706	13.211	9.603	126.860	27
28	5.112	.1956	.0146	.0746	68.528	13.406	9.857	132.142	28
29	5.418	.1846	.0136	.0736	73.640	13.591	10.103	137.309	29
30	5.743	.1741	.0126	.0726	79.058	13.765	10.342	142.359	30

1C) A machine shop plans to buy a CNC (Computer Numerical Controlled) mill to automate simple steps for large-scale production jobs, saving labor costs. The mill costs ₹10,00,000. Using the mill would result in the following labor cost savings:

Year	1	2	3	4	5	6
Savings (₹)	2,80,000	2,60,000	2,40,000	2,20,000	2,00,000	1,80,000

Should the mill be purchased if the company's discount rate is 10%? (3)

10% Compound Interest Factors 10%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0	1
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826	2
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329	3
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378	4
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862	5
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684	6
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763	7
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029	8
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421	9
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891	10
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396	11
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901	12
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377	13
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801	14
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152	15

10% Compound Interest Factors 10%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
16	4.595	.2176	.0278	.1278	35.950	7.824	5.549	43.416	16
17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582	17
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640	18
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583	19
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407	20
21	7.400	.1351	.0156	.1156	64.003	8.649	6.719	58.110	21
22	8.140	.1228	.0140	.1140	71.403	8.772	6.919	60.689	22
23	8.954	.1117	.0126	.1126	79.543	8.883	7.108	63.146	23
24	9.850	.1015	.0113	.1113	88.497	8.985	7.288	65.481	24
25	10.835	.0923	.0102	.1102	98.347	9.077	7.458	67.696	25
26	11.918	.0839	.00916	.1092	109.182	9.161	7.619	69.794	26
27	13.110	.0763	.00826	.1083	121.100	9.237	7.770	71.777	27
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650	28
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415	29
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077	30

2A) A company acquired specialized equipment for $1,950, anticipating it would retain a salvage value of $450 after a 5-year useful life. To analyse depreciation impact on financial statements, calculate and present a detailed depreciation schedule using the following two approaches: (4)
a) If the company adopts the **Straight-Line depreciation method**.
b) If the company uses the **Double Declining Balance method**.

2B) An industrial firm is considering two processes for manufacturing storm drains. (3)
Plan 1 (conventional injection moulding): It requires a initial cost of $2,000,000. The cost for inspecting, maintaining, and cleaning the moulds is expected to be $ 6000 per year. The cost of materials for this plan is expected to be the same as for the other plan.
Plan 2 (virtual engineered composites): The first cost to tool the floating mould is only $27,000. However, because of the newness of the process, personnel and product-reject costs are expected to be higher than those for a conventional process. The company expects the operating costs to be $46,000 per year for the first 8 years and then to decrease to $11,000 per year thereafter.
At an interest rate of 10% per year, which process should the company select based on present worth analysis over a 14-year study period?

10% Compound Interest Factors 10%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0	1
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826	2
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329	3
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378	4
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862	5
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684	6
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763	7
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029	8
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421	9
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891	10
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396	11
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901	12
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377	13
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801	14
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152	15

10% Compound Interest Factors 10%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
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17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582	17
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640	18
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583	19
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407	20
21	7.400	.1351	.0156	.1156	64.003	8.649	6.719	58.110	21
22	8.140	.1228	.0140	.1140	71.403	8.772	6.919	60.689	22
23	8.954	.1117	.0126	.1126	79.543	8.883	7.108	63.146	23
24	9.850	.1015	.0113	.1113	88.497	8.985	7.288	65.481	24
25	10.835	.0923	.0102	.1102	98.347	9.077	7.458	67.696	25
26	11.918	.0839	.00916	.1092	109.182	9.161	7.619	69.794	26
27	13.110	.0763	.00826	.1083	121.100	9.237	7.770	71.777	27
28	14.421	.0693	.00745	.1075	134.210	9.307	7.914	73.650	28
29	15.863	.0630	.00673	.1067	148.631	9.370	8.049	75.415	29
30	17.449	.0573	.00608	.1061	164.494	9.427	8.176	77.077	30

2C) A hospital plans to install a new air purification system with an initial setup cost of $300,000. Maintenance costs begin at $25,000 in year 3, increasing by $4,000 annually until year 7. After year 7, the maintenance cost remains constant at the year-7 level indefinitely. Operating costs start at $35,000 per year from year 1 and remain constant over time. Additionally, the system requires a major component replacement every 12 years at a cost of $120,000. Given an annual interest rate of 6%, calculate the capitalized cost of the system.

6% Compound Interest Factors 6%

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n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
1	1.060	.9434	1.0000	1.0600	1.000	0.943	0	0	1
2	1.124	.8900	.4854	.5454	2.060	1.833	0.485	0.890	2
3	1.191	.8396	.3141	.3741	3.184	2.673	0.961	2.569	3
4	1.262	.7921	.2286	.2886	4.375	3.465	1.427	4.945	4
5	1.338	.7473	.1774	.2374	5.637	4.212	1.884	7.934	5
6	1.419	.7050	.1434	.2034	6.975	4.917	2.330	11.459	6
7	1.504	.6651	.1191	.1791	8.394	5.582	2.768	15.450	7
8	1.594	.6274	.1010	.1610	9.897	6.210	3.195	19.841	8
9	1.689	.5919	.0870	.1470	11.491	6.802	3.613	24.577	9
10	1.791	.5584	.0759	.1359	13.181	7.360	4.022	29.602	10
11	1.898	.5268	.0668	.1268	14.972	7.887	4.421	34.870	11
12	2.012	.4970	.0593	.1193	16.870	8.384	4.811	40.337	12
13	2.133	.4688	.0530	.1130	18.882	8.853	5.192	45.963	13
14	2.261	.4423	.0476	.1076	21.015	9.295	5.564	51.713	14
15	2.397	.4173	.0430	.1030	23.276	9.712	5.926	57.554	15

(3)

6% Compound Interest Factors 6%

	Single Payment		Uniform Payment Series				Arithmetic Gradient		
n	Compound Amount Factor Find F Given P F/P	Present Worth Factor Find P Given F P/F	Sinking Fund Factor Find A Given F A/F	Capital Recovery Factor Find A Given P A/P	Compound Amount Factor Find F Given A F/A	Present Worth Factor Find P Given A P/A	Gradient Uniform Series Find A Given G A/G	Gradient Present Worth Find P Given G P/G	n
16	2.540	.3936	.0390	.0990	25.672	10.106	6.279	63.459	16
17	2.693	.3714	.0354	.0954	28.213	10.477	6.624	69.401	17
18	2.854	.3503	.0324	.0924	30.906	10.828	6.960	75.357	18
19	3.026	.3305	.0296	.0896	33.760	11.158	7.287	81.306	19
20	3.207	.3118	.0272	.0872	36.786	11.470	7.605	87.230	20
21	3.400	.2942	.0250	.0850	39.993	11.764	7.915	93.113	21
22	3.604	.2775	.0230	.0830	43.392	12.042	8.217	98.941	22
23	3.820	.2618	.0213	.0813	46.996	12.303	8.510	104.700	23
24	4.049	.2470	.0197	.0797	50.815	12.550	8.795	110.381	24
25	4.292	.2330	.0182	.0782	54.864	12.783	9.072	115.973	25
26	4.549	.2198	.0169	.0769	59.156	13.003	9.341	121.468	26
27	4.822	.2074	.0157	.0757	63.706	13.211	9.603	126.860	27
28	5.112	.1956	.0146	.0746	68.528	13.406	9.857	132.142	28
29	5.418	.1846	.0136	.0736	73.640	13.591	10.103	137.309	29
30	5.743	.1741	.0126	.0726	79.058	13.765	10.342	142.359	30

3A) (4)

A school is evaluating two different heating systems for its gymnasium with MARR of 10%: an electric heater (Alternative A) and a gas furnace (Alternative B).

	Alternative A Electric Heater	Alternative B Gas Furnace
Initial Cost:	$8,000	$6,000
Salvage Value	$1,500	$1,000
Maintenance Cost	$400 per year	$200 per year
Operating Cost	$0.20 per kWh	$0.20 per kWh
Estimated useful life:	10 years	10 years

Requirements:

a) Calculate the break-even number of kWh per year (K) at which the total costs for both alternatives are equal.

b) If the estimated operation is 20,000 kWh per year, determine which alternative is more economical.

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901

3B) Fulton National Hospital is reviewing ways of cutting the cost of stocking medical supplies. Two new stockless systems are being considered, to lower the hospital's holding and handling costs. The hospital's industrial engineer has compiled the relevant financial data for each system as follows

	Current Practice	Just in time	Stockless supply system
Startup cost	$0	$2.5	$5
Annual stock holding cost	$3	$1.4	$0.2
Annual operating cost	$2	$1.5	$1.2
System life (years)	8	8	8
			Note: dollar values are in millions

The system life of eight years represents the period that the contract with the medical suppliers is in force. Using Incremental IRR method of analysis, determine which system is more economical? The hospital's MARR is 10%. (4)

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
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8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
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10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901

3C) A company is evaluating an alternative for outsourcing its IT infrastructure management over a 6-year period. The details of alternative are as follows:

	Alternative A
Initial setup cost	$80,000
Maintenance cost (on semi-annual basis)	$12,000 at the end of the first year and then onwards increasing by $2,000 every semi-annual
System upgrade cost at year 3	$30,000

(2)

Determine the annual worth of alternative A using an interest rate is 16%, compounded semi-annually.

4A) Agod Electrical Ltd. is considering of replacing its old diesel generator system with a new electric generator system. The old system was bought 8 years ago with an expected life of 16 years at time of purchase. The other details of these two systems are provided below:

	Old Generator	New Generator
Initial Cost (at time of purchase)	₹2,20,000	₹2,00,000
Annual Operating Cost (including maintenance, fuel, etc.)	₹16,000	₹8,000
Estimated Life at time of purchase	16 years	10 years
Estimated Salvage Value	₹24,000	₹40,000

Assume that the old generator can be sold now at book value in the market and the company uses straight line method of depreciation.
Using an appropriate replacement analysis approach with MARR of 10%, determine the best course of action for the company.

(4)

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891
11	2.853	.3505	.0540	.1540	18.531	6.495	4.064	26.396
12	3.138	.3186	.0468	.1468	21.384	6.814	4.388	29.901
13	3.452	.2897	.0408	.1408	24.523	7.103	4.699	33.377
14	3.797	.2633	.0357	.1357	27.975	7.367	4.996	36.801
15	4.177	.2394	.0315	.1315	31.772	7.606	5.279	40.152
16	4.595	.2176	.0278	.1278	35.950	7.824	5.549	43.416
17	5.054	.1978	.0247	.1247	40.545	8.022	5.807	46.582
18	5.560	.1799	.0219	.1219	45.599	8.201	6.053	49.640
19	6.116	.1635	.0195	.1195	51.159	8.365	6.286	52.583
20	6.728	.1486	.0175	.1175	57.275	8.514	6.508	55.407

4B) (4)

The project has an estimated life of 5 years, and the cash flows are uncertain. The company has identified three possible scenarios: optimistic, most likely, and pessimistic. The expected cash flows for each scenario are as follows:

Optimistic Scenario:

	Year 1	Year 2	Year 3	Year 4	Year 5
Cashflow	$300,000	$400,000	$500,000	$600,000	$700,000

Most Likely Scenario:

	Year 1	Year 2	Year 3	Year 4	Year 5
Cashflow	$250,000	$350,000	$450,000	$550,000	$650,000

Pessimistic Scenario:

	Year 1	Year 2	Year 3	Year 4	Year 5
Cashflow	$200,000	$300,000	$400,000	$500,000	$600,000

Assuming a discount rate of 10%, answer the following questions:

a) Calculate the Net Present Value (NPV) for each scenario.

b) Discuss how scenario analysis helps in understanding the risk associated with this investment.

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891

4C) A retail company, RetailCo, has recently reported a current ratio of 1.2 and a quick ratio of 0.8. Some members of the management team argue that these liquidity ratios indicate a healthy financial position, while others express concern that the quick ratio, being below 1, suggests potential liquidity issues. Justify whether RetailCo's liquidity ratios indicate a strong or weak liquidity position. (2)

5A) (3)

You work for Midstates Solar Power. A manager asked you to determine which of the following two machines will have the lower

(*a*) capital recovery and

(*b*) equivalent annual total cost.

Machine Semi2 has a first cost of $80,000 and an operating cost of $21,000 in year 1, increasing by $500 per year through year 5, after which time it will have a salvage value of $13,000.

Machine Auto1 has a first cost of $62,000 and an operating cost of $21,000 in year 1, increasing by a gradient of $1,700 per year through year 5, after which time it will have a salvage value of $2000.

Utilize an interest rate of 10% per year to determine both the estimates asked above.

10%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.100	.9091	1.0000	1.1000	1.000	0.909	0	0
2	1.210	.8264	.4762	.5762	2.100	1.736	0.476	0.826
3	1.331	.7513	.3021	.4021	3.310	2.487	0.937	2.329
4	1.464	.6830	.2155	.3155	4.641	3.170	1.381	4.378
5	1.611	.6209	.1638	.2638	6.105	3.791	1.810	6.862
6	1.772	.5645	.1296	.2296	7.716	4.355	2.224	9.684
7	1.949	.5132	.1054	.2054	9.487	4.868	2.622	12.763
8	2.144	.4665	.0874	.1874	11.436	5.335	3.004	16.029
9	2.358	.4241	.0736	.1736	13.579	5.759	3.372	19.421
10	2.594	.3855	.0627	.1627	15.937	6.145	3.725	22.891

5B) XYZ Corp is a medium-sized manufacturing company seeking to analyze its financial stability and ability to meet its long-term obligations. For the fiscal year ending December 31, 2023, XYZ Corp reported the following financial data:

- Total Assets: $3,000,000
- Total Liabilities: $1,800,000
- Shareholders' Equity: $1,200,000
- EBIT (Earnings Before Interest and Taxes): $500,000
- Interest Expense: $100,000

(3)

Calculate the interest coverage ratio, debt-equity ratio, and debt ratio for XYZ Corp and discuss their significance for the company.

5C) A piece of machinery costs $7,500 and has no salvage value after it is installed. The manufacturer's warranty will pay the first year's maintenance and repair costs. In the

8%

n	F/P	P/F	A/F	A/P	F/A	P/A	A/G	P/G
1	1.080	.9259	1.0000	1.0800	1.000	0.926	0	0
2	1.166	.8573	.4808	.5608	2.080	1.783	0.481	0.857
3	1.260	.7938	.3080	.3880	3.246	2.577	0.949	2.445
4	1.360	.7350	.2219	.3019	4.506	3.312	1.404	4.650
5	1.469	.6806	.1705	.2505	5.867	3.993	1.846	7.372
6	1.587	.6302	.1363	.2163	7.336	4.623	2.276	10.523
7	1.714	.5835	.1121	.1921	8.923	5.206	2.694	14.024
8	1.851	.5403	.0940	.1740	10.637	5.747	3.099	17.806

(4)

second year, maintenance costs will be $900, and this item will increase on a $900 arithmetic gradient in subsequent years. Also, operating expenses for the machinery will be $500 in the first year and will increase on a $400 arithmetic gradient in the following years. If interest is 8%, what is the economic service life for this machine?

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